# Parameter Overlay Module (POM)

This chapter describes the parameter overlay module (POM).

**Topic** **Page**

## 18.1 Introduction

In many applications it is important to be able to change certain parameters in the program without having to re-flash the device and immediately test these changes either in a hardware-in-the-loop simulation or in a real environment.

The POM provides a mechanism to redirect accesses to non-volatile memory into a volatile memory internal or external to the device. The data requested by the CPU will be fetched from the overlay memory instead of the main non-volatile memory.

### *18.1.1 Main Features*

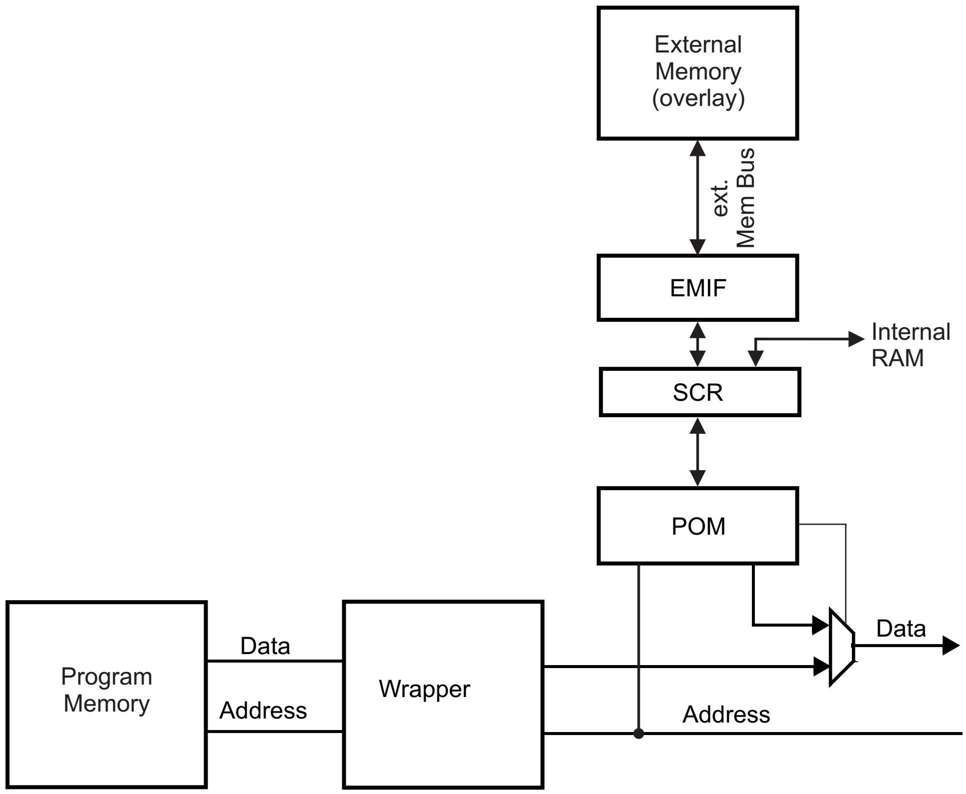
- Redirects program memory accesses to internal or external memory (overlay memory)
- Up to 8 MByte of external overlay memory
- Provides up to 32 programmable memory regions to replace non-volatile memory
  - Programmable region start address
  - Programmable region size (64 Bytes up to 256 kBytes in power of 2 steps)

### *18.1.2 Parameter Overlay Module (POM) Considerations*

- The POM can map onto up to 8MB of the internal or external memory space. The starting address and the size of the memory overlay are configurable through the POM control registers. Care must be taken to ensure that the overlay is mapped on to available memory.
- ECC must be disabled by software through CP15 in case POM overlay is enabled; otherwise ECC errors will be generated.
- POM overlay must not be enabled when the flash and internal RAM memories are swapped through the MEM SWAP field of the Bus Matrix Module Control Register 1 (BMMCR1).
- When POM is used to overlay the flash on to internal or external RAM, there is a bus contention possibility when another master accesses the TCM flash. This results in a system hang.
  - The POM implements a timeout feature to detect this exact scenario. The timeout needs to be enabled whenever POM overlay is enabled.
  - The timeout can be enabled by writing Ah to the enable timeout (ETO) field of the POM global control register (POMGLBCTRL).
  - In case a read request by the POM cannot be completed within 32 HCLK cycles, the timeout (TO) flag is set in the POM status register (POMFLG). Also, an abort is generated to the CPU. This can be a prefetch abort for an instruction fetch or a data abort for a data fetch.
  - The prefetch-abort and data-abort handlers must be modified to check if the TO flag in the POM is set. If so, then the application can assume that the timeout is caused by a bus contention between the POM transaction and another master accessing the same memory region. The abort handlers need to clear the TO flag, so that any further aborts are not misinterpreted as having been caused due to a timeout from the POM.

### 18.1.3 Block Diagram

**Figure 18-1. System Overlay Block Diagram**




## 18.2 Module Operation

The POM has up to 32 programmable regions. Whenever the CPU requests data from the non-volatile memory which address falls into one of the programmed regions, the POM will request the data from the overlay memory. Wait states will be automatically inserted until the data is available from the overlay memory. This ensures that the overlay memory has the same (in the case that the latency from overlay memory is less than the program memory latency) or slower access time than the program memory.

The POM does not provide a feature to write into the overlay memory. The write has to be performed directly to the memory mapped address space of the overlay memory.

When the module is disabled, no redirection of access will be performed.

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**NOTE:** The overlay feature is not available for any other bus masters other than the main CPU. Any attempt to access overlaid memory via the POM by any other bus master would result in a deadlock situation (system hang). Other bus masters need to access the target internal or external memory directly.

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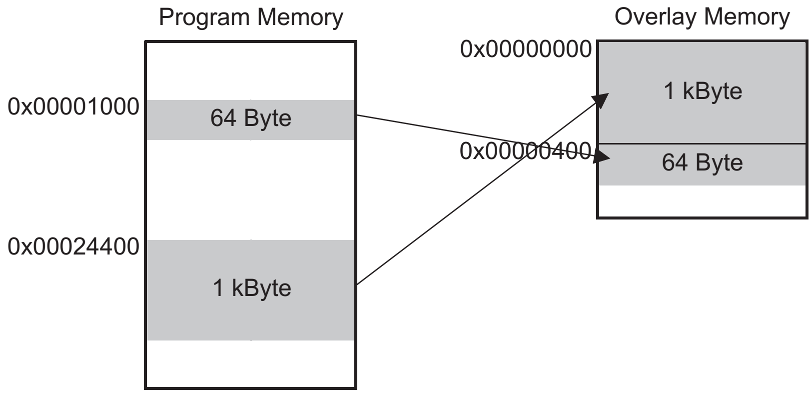
### 18.2.1 Decode Regions

There are 32 decode regions. Regions are defined by a start address and a region size. The start address is 23 bits wide to cover the maximum 8 MByte address space defined for program memory. The region size ranges from 64 Bytes to 256 kBytes with a step size of power of 2. If a region size of 0 is selected, the region is disabled. To support overlay memory that is not the same size as the program memory, an overlay start address that is 23-bits wide can be specified. The size of the overlay region is the same as the program memory region. The start address of both program memory and overlay memory region have to be a multiple of the programmed region size. When the address of the access falls into one of the programmed regions, the POM will start requesting the data from the overlay memory address, based on the overlay start address. If the address falls into multiple overlapping regions, the region with the lowest number has highest priority and only one read request from overlay memory will be initiated. This avoids that regions with the same program memory address, but different overlay memory addresses, request different data.

**Figure 18-2. Region Definition Example**




### 18.2.2 Bus Errors on Accesses via POM

In case a read by the POM cannot be completed within 32 HCLK cycles due to other transactions ongoing or due to other bus errors, a timeout mechanism is implemented. This timeout mechanism is disabled by default and must be enabled by writing Ah to the enable timeout (ETO) field of the POM global control register (POMGLBCTRL).

An abort is generated to the host CPU when this timeout occurs. The host CPU responds by either taking the prefetch abort or the data abort exceptions depending on the nature of the access that caused the timeout.

The abort handler can check if the TO flag is set in the POM module's flag register (POMFLG). If this flag is set, then the application can assume that the timeout is caused by the POM access not completing. This flag must be cleared so that any further aborts are not misinterpreted as having been caused by timeout on a POM access.

## 18.3 POM Control Registers

Table 18-1 lists the Parameter Overlay Module registers. The registers support only 32-bit writes. The offset is relative to the associated peripheral select. The base address for the control packets is FFA0 4000h.

**Table 18-1. POM Registers**

| Offset | Acronym | Register Description | Section |
|---|---|---|---|
| 00h | POMGLBCTRL | POM Global Control Register | Section 18.3.1 |
| 04h | POMREV | POM Revision ID | Section 18.3.2 |
| 08h | POMCLKCTRL | POM Clock Gate Control Register | Section 18.3.3 |
| 0Ch | POMFLG | POM Status Register | Section 18.3.4 |
| 200h, 210h, ..., 3F0h | POMPROGSTARTx | POM Program Region Start Address Register x | Section 18.3.5 |
| 204h, 214h, ..., 3F4h | POMOVLSTARTx | POM Overlay Region Start Address Register x | Section 18.3.6 |
| 208h, 218h, ..., 3F8h | POMREGSIZEx | POM Region Size Register x | Section 18.3.7 |
| F00h | POMITCTRL | POM Integration Control Register | Section 18.3.8 |
| FA0h | POMCLAIMSET | POM Claim Set Register | Section 18.3.9 |
| FA4h | POMCLAIMCLR | POM Claim Clear Register | Section 18.3.10 |
| FB0h | POMLOCKACCESS | POM Lock Access Register | Section 18.3.11 |
| FB4h | POMLOCKSTATUS | POM Lock Status Register | Section 18.3.12 |
| FB8h | POMAUTHSTATUS | POM Authentication Status Register | Section 18.3.13 |
| FC8h | POMDEVID | POM Device ID Register | Section 18.3.14 |
| FCCh | POMDEVTYPE | POM Device Type Register | Section 18.3.15 |
| FD0h | POMPERIPHERALID4 | POM Peripheral ID 4 Register | Section 18.3.16 |
| FD4h | POMPERIPHERALID5 | POM Peripheral ID 5 Register | Section 18.3.17 |
| FD8h | POMPERIPHERALID6 | POM Peripheral ID 6 Register | Section 18.3.18 |
| FDCh | POMPERIPHERALID7 | POM Peripheral ID 7 Register | Section 18.3.19 |
| FE0h | POMPERIPHERALID0 | POM Peripheral ID 0 Register | Section 18.3.20 |
| FE4h | POMPERIPHERALID1 | POM Peripheral ID 1 Register | Section 18.3.21 |
| FE8h | POMPERIPHERALID2 | POM Peripheral ID 2 Register | Section 18.3.22 |
| FECh | POMPERIPHERALID3 | POM Peripheral ID 3 Register | Section 18.3.23 |
| FF0h | POMCOMPONENTID0 | POM Component ID 0 Register | Section 18.3.24 |
| FF4h | POMCOMPONENTID1 | POM Component ID 1 Register | Section 18.3.25 |
| FF8h | POMCOMPONENTID2 | POM Component ID 2 Register | Section 18.3.26 |
| FFCh | POMCOMPONENTID3 | POM Component ID 3 Register | Section 18.3.27 |

### 18.3.1 POM Global Control Register (POMGLBCTRL)

This register contains a key to enable the POM module. Logic remains reset until this key is set.

**Figure 18-3. POM Global Control Register (POMGLBCTRL) [address = FFA0 4000h]**

| 31–23 | 22–16 |
|---|---|
| OTADDR | Reserved |
| R/WP-60h | R-0 |

| 15–12 | 11–8 | 7–4 | 3–0 |
|---|---|---|---|
| Reserved | ETO | Reserved | ON/OFF |
| R-0 | R/WP-5h | R-0 | R/WP-5h |

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privilege mode only; -*n* = value after reset

**Table 18-2. POM Global Control Register (POMGLBCTRL) Field Descriptions**

| Bit | Field | Value | Description |
|---|---|---|---|
| 31-23 | OTADDR | 60h | **Overlay target Address**.<br>These bits determine the upper address bits of the target overlay address. Writing a different value to this field will steer the POM access to a different location in the 4GB address space. The application must ensure that the overlay memory address points to a valid internal or external memory location. |
| 22-12 | Reserved | 0 | Reads return 0, writes have no effect. |
| 11-8 | ETO | | **Enable Timeout**. Refer to Section 18.2.2 for more details on the timeout error. |
| | | Ah | Timeout for bus transactions is enabled. |
| | | All other values | Timeout for bus transactions is disabled. The timeout is disabled by default. |
| 7-4 | Reserved | 0 | Reads return zeros, writes have no effect. |
| 3-0 | ON/OFF | | **Turn functionality of POM on or off.** |
| | | Ah | POM is functional. |
| | | All other values | POM is held in reset. |
| | | | NOTE: The key should be written to 5h, to avoid single bit flips inadvertently turning on the module. |

### 18.3.2 POM Revision ID (POMREV)

This register contains the revision ID of the POM module.

**Figure 18-4. POM Revision ID (POMREV) [address = FFA0 4004h]**

| 31-30 | 29-28 | 27-16 |
|---|---|---|
| SCHEME | Reserved | FUNC |
| R-1h | R-0 | R-A03h |

| 15-11 | 10-8 | 7-6 | 5-0 |
|---|---|---|---|
| RTL | MAJOR | CUSTOM | MINOR |
| R-0 | R-1h | R-0 | R-8h |

LEGEND: R/W = Read/Write; R = Read only; -*n* = value after reset

**Table 18-3. POM Revision ID (POMREV) Field Descriptions**

| Bit | Field | Value | Description |
|---|---|---|---|
| 31-30 | SCHEME | 1h | Used to distinguish between different ID schemes. |
| 29-28 | Reserved | 0 | Reads return 0, writes have no effect. |
| 27-16 | FUNC | A03h | Indicates the SW compatible module family. |
| 15-11 | RTL | 0 | RTL version number. |
| 10-8 | MAJOR | 1h | Major revision number. |
| 7-6 | CUSTOM | 0 | Indicates a device specific implementation. |
| 5-0 | MINOR | 8h | Minor revision number. |

### 18.3.3 POM Clock Gate Control Register (POMCLKCTRL)

This register is for TI internal use only.

**Figure 18-5. POM Clock Gate Control Register [address = FFA0 4008h]**

| 31-16 |
|---|
| Reserved |
| R-0 |

| 15-1 | 0 |
|---|---|
| Reserved | CLK GATE OFF |
| R-0 | R/W-0 |

LEGEND: R/W = Read/Write; R = Read only; -*n* = value after reset

**Table 18-4. POM Clock Gate Control Register (POMCLKCTRL) Field Descriptions**

| Bit | Field | Value | Description |
|---|---|---|---|
| 31-1 | Reserved | 0 | Reads return 0, writes have no effect. |
| 0 | CLK GATE OFF | 0 | Do not modify this bit. Leave it in its reset state. Modifying the bit while the POM module is switched on can result in unexpected behavior. |

### 18.3.4 POM Status Register (POMFLG)

This register provides POM status information.

**Figure 18-6. POM Status Register [address = FFA0 400Ch]**

| 31–16 |
|---|
| Reserved |
| R-0 |

| 15–1 | 0 |
|---|---|
| Reserved | TO |
| R-0 | R/W1CP-0 |

LEGEND: R/W = Read/Write; R = Read only; W1CP = Write 1 to clear in privilege mode only; -*n* = value after reset

**Table 18-5. POM Status Register (POMFLG) Field Descriptions**

| Bit | Field | Value | Description |
|---|---|---|---|
| 31-1 | Reserved | 0 | Reads return 0, writes have no effect. |
| 0 | TO | | **Timeout**.<br>This flag signals a timeout condition on the last CPU read access through the POM. This flag is only updated when the POM module is enabled (ON/OFF = Ah). |
| | | 0 | Read: No timeout occurred.<br>Write: No effect. |
| | | 1 | Read: Timeout occurred.<br>Write: Bit is cleared. |

### 18.3.5 POM Program Region Start Address Register x (POMPROGSTARTx)

This register contains the start address of the region in program memory. x goes from 0 to 31, since up to 32 regions can be defined.

**Figure 18-7. POM Program Region Start Register x (POMPROGSTARTx) [address = FFA0 4200h, FFA0 4210h, ..., FFA0 43F0h]**

| 31–23 | 22–16 |
|---|---|
| Reserved | STARTADDRESS |
| R-0 | R/WP-0 |

| 15–0 |
|---|
| STARTADDRESS |
| R/WP-0 |

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privilege mode only; -*n* = value after reset

**Table 18-6. POM Program Region Start Address Register x (POMPROGSTARTx) Field Descriptions**

| Bit | Field | Value | Description |
|---|---|---|---|
| 31-23 | Reserved | 0 | Reads return 0, writes have no effect. |
| 22-0 | STARTADDRESS | | Defines the start address of the program memory region. The start address has to be a multiple of the region size. |

**NOTE:** If the region start address is programmed to a non-region size boundary, the region will begin at the next lower region size boundary.

### 18.3.6 POM Overlay Region Start Address Register x (POMOVLSTARTx)

This register contains the start address of the region in overlay memory. x goes from 0 to 31, since up to 32 regions can be defined.

**Figure 18-8. POM Overlay Region Start Register x (POMOVLSTARTx) [address = FFA0 4204h, FFA0 4214h, ..., FFA0 43F4h]**

| 31–23 | 22–16 |
|---|---|
| Reserved | STARTADDRESS |
| R-0 | R/W-0 |

| 15–0 |
|---|
| STARTADDRESS |
| R/W-0 |

LEGEND: R/W = Read/Write; R = Read only; -*n* = value after reset

**Table 18-7. POM Overlay Region Start Address Register x (POMOVLSTARTx) Field Descriptions**

| Bit | Field | Value | Description |
|---|---|---|---|
| 31-23 | Reserved | 0 | Reads return 0, writes have no effect. |
| 22-0 | STARTADDRESS | | Defines the start address of the overlay memory region. The start address has to be a multiple of the region size. |

**NOTE:** If the region start address is programmed to a non-region size boundary, the region will begin at the next lower region size boundary.

### 18.3.7 POM Region Size Register x (POMREGSIZEx)

This register contains the region size for both program memory and overlay memory. x goes from 0 to 31, since up to 32 regions can be defined.

**Figure 18-9. POM Region Size Register x (POMREGSIZEx) [address = FFA0 4208h, FFA0 4218h, ..., FFA0 43F8h]**

| 31–16 |
|---|
| Reserved |
| R-0 |

| 15–4 | 3–0 |
|---|---|
| Reserved | SIZE |
| R-0 | R/W-0 |

LEGEND: R/W = Read/Write; R = Read only; -*n* = value after reset

**Table 18-8. POM Region Size Register x (POMREGSIZEx) Field Descriptions**

| Bit | Field | Value | Description |
|---|---|---|---|
| 31-4 | Reserved | 0 | Reads return 0, writes have no effect. |
| 3-0 | SIZE | | **Region size** |
| | | 0 | Region is disabled. |
| | | 1h | 64 Bytes |
| | | 2h | 128 Bytes |
| | | : | : |
| | | Dh | 256 kBytes |
| | | Eh-Fh | Reserved |

**NOTE:** If the region is enabled by writing a non-zero value to the SIZE bitfield, it will take some number of VCLK cycles until the write takes effect. If during this time an access to the programmed region is taking place and the POM is already enabled in the POMGLBCTRL register, it either decodes the previous setting of the SIZE field or the access will be directed to the program memory when the region was disabled.

### 18.3.8 POM Integration Control Register (POMITCTRL)

This is a CoreSight register and is for debug purpose only. The integration functionality is not implemented. The register reads 00000000.

**Figure 18-10. POM Integration Control Register (POMITCTRL) [address = FFA0 4F00h]**

| 31–0 |
|---|
| Reserved |
| R-0 |

LEGEND: R = Read only; -*n* = value after reset

**Table 18-9. POM Integration Control Register (POMITCTRL) Field Descriptions**

| Bit | Field | Value | Description |
|---|---|---|---|
| 31-0 | Reserved | 0 | Reads return 0, writes have no effect. |

### *18.3.9 POM Claim Set Register (POMCLAIMSET)*

This is a CoreSight register and is for debug purpose only. This register allows different masters to claim the control for the POM module. There is no control functionality for POM register accesses implemented. The only functionality of these bits is to indicate that another master is controlling the module.

**Figure 18-11. POM Claim Set Register (POMCLAIMSET) [address = FFA0 4FA0h]**

| 31–16 |
|---|
| Reserved |
| R-0 |

| 15–2 | 1 | 0 |
|---|---|---|
| Reserved | SET1 | SET0 |
| R-0 | R/W-1 | R/W-1 |

LEGEND: R/W = Read/Write; R = Read only; -*n* = value after reset

**Table 18-10. POM Claim Set Register (POMCLAIMSET) Field Descriptions**

| Bit | Field | Value | Description |
|---|---|---|---|
| 31-2 | Reserved | 0 | Reads return 0, writes have no effect. |
| 1 | SET1 | | The module is claimed. |
| | | 0 | Read: This claim tag bit is not implemented.<br>Write: No effect. |
| | | 1 | Read: This claim tag is implemented.<br>Write: Set claim tag. |
| 0 | SET0 | | The module is claimed. |
| | | 0 | Read: This claim tag bit is not implemented.<br>Write: No effect. |
| | | 1 | Read: This claim tag is implemented.<br>Write: Set claim tag. |

### 18.3.10 POM Claim Clear Register (POMCLAIMCLR)

This is a CoreSight register and is for debug purpose only. This register allows different masters to claim the control for the POM module. There is no control functionality for POM register accesses implemented. The only functionality of these bits is to indicate that another master is controlling the module.

**Figure 18-12. POM Claim Clear Register (POMCLAIMCLR) [address = FFA0 4FA4h]**

| 31–16 |
|---|
| Reserved |
| R-0 |

| 15–2 | 1 | 0 |
|---|---|---|
| Reserved | CLR1 | CLR0 |
| R-0 | R/W-1 | R/W-1 |

LEGEND: R/W = Read/Write; R = Read only; -*n* = value after reset

**Table 18-11. POM Claim Clear Register (POMCLAIMCLR) Field Descriptions**

| Bit | Field | Value | Description |
|---|---|---|---|
| 31-2 | Reserved | 0 | Reads return 0, writes have no effect. |
| 1 | CLR1 | | The module is claimed. |
| | | | Read: Current claim tag value. |
| | | 0 | Write: No effect. |
| | | 1 | Write: Clear claim tag. |
| 0 | CLR0 | | The module is claimed. |
| | | | Read: Current claim tag value. |
| | | 0 | Write: No effect. |
| | | 1 | Write: Clear claim tag. |

### 18.3.11 POM Lock Access Register (POMLOCKACCESS)

This is a CoreSight register and is for debug purpose only. The register reads 00000000.

**Figure 18-13. POM Lock Access Register (POMLOCKACCESS) [address = FFA0 4FB0h]**

| 31 — 0 |
|---|
| Reserved |
| R-0 |

LEGEND: R = Read only; -*n* = value after reset

**Table 18-12. POM Lock Access Register (POMLOCKACCESS) Field Descriptions**

| Bit | Field | Value | Description |
|---|---|---|---|
| 31-0 | Reserved | 0 | Reads return 0, writes have no effect. |

### 18.3.12 POM Lock Status Register (POMLOCKSTATUS)

This is a CoreSight register and is for debug purpose only. The register reads 00000000.

**Figure 18-14. POM Lock Status Register (POMLOCKSTATUS) [address = FFA0 4FB4h]**

| 31 — 0 |
|---|
| Reserved |
| R-0 |

LEGEND: R = Read only; -*n* = value after reset

**Table 18-13. POM Lock Status Register (POMLOCKSTATUS) Field Descriptions**

| Bit | Field | Value | Description |
|---|---|---|---|
| 31-0 | Reserved | 0 | Reads return 0, writes have no effect. |

### 18.3.13 POM Authentication Status Register (POMAUTHSTATUS)

This is a CoreSight register and is for debug purpose only. The register reads 00000000.

**Figure 18-15. POM Authentication Status Register (POMAUTHSTATUS) [address = FFA0 4FB8h]**

| 31 — 0 |
|---|
| Reserved |
| R-0 |

LEGEND: R = Read only; -*n* = value after reset

**Table 18-14. POM Authentication Status Register (POMAUTHSTATUS) Field Descriptions**

| Bit | Field | Value | Description |
|---|---|---|---|
| 31-0 | Reserved | 0 | Reads return 0, writes have no effect. |

### 18.3.14 POM Device ID Register (POMDEVID)

This is a CoreSight register and is for debug purpose only. The register reads 00000000.

**Figure 18-16. POM Device ID Register (POMDEVID) [address = FFA0 4FC8h]**

| 31–0 |
|---|
| Reserved |
| R-0 |

LEGEND: R = Read only; -*n* = value after reset

**Table 18-15. POM Device ID Register (POMDEVID) Field Descriptions**

| Bit | Field | Value | Description |
|---|---|---|---|
| 31-0 | Reserved | 0 | Reads return 0, writes have no effect. |

### 18.3.15 POM Device Type Register (POMDEVTYPE)

This is a CoreSight register and is for debug purpose only. The device type register defines the CoreSight module class.

**Figure 18-17. POM Device Type Register (POMDEVTYPE) [address = FFA0 4FCCh]**

| 31–16 |
|---|
| Reserved |
| R-0 |

| 15–8 | 7–4 | 3–0 |
|---|---|---|
| Reserved | Sub Type | Major Type |
| R-0 | R-0 | R-4h |

LEGEND: R/W = Read/Write; R = Read only; -*n* = value after reset

**Table 18-16. POM Device Type Register (POMDEVTYPE) Field Descriptions**

| Bit | Field | Value | Description |
|---|---|---|---|
| 31-8 | Reserved | 0 | Reads return 0, writes have no effect. |
| 7-4 | Sub Type | 0 | Other |
| 3-0 | Major Type | 4h | Debug Control |

### 18.3.16 POM Peripheral ID 4 Register (POMPERIPHERALID4)

This is a CoreSight register and is for debug purpose only. This register shows the peripheral identification of the CoreSight component.

**Figure 18-18. POM Peripheral ID 4 Register (POMPERIPHERALID4) [address = FFA0 4FD0h]**

| 31–16 |
|---|
| Reserved |
| R-0 |

| 15–8 | 7–4 | 3–0 |
|---|---|---|
| Reserved | 4KB Count | JEP106 Continuation Code |
| R-0 | R-0 | R-0 |

LEGEND: R/W = Read/Write; R = Read only; -*n* = value after reset

**Table 18-17. POM Peripheral ID 4 Register (POMPERIPHERALID4) Field Descriptions**

| Bit | Field | Value | Description |
|---|---|---|---|
| 31-8 | Reserved | 0 | Reads return 0, writes have no effect. |
| 7-4 | 4KB Count | 0 | Only 4KB is implemented. |
| 3-0 | JEP Continuation Code | 0 | JEP106 Code |

### 18.3.17 POM Peripheral ID 5 Register (POMPERIPHERALID5)

This is a CoreSight register and is for debug purpose only. This register reads 00000000.

**Figure 18-19. POM Peripheral ID 5 Register (POMPERIPHERALID5) [address = FFA0 4FD4h]**

| 31–0 |
|---|
| Reserved |
| R-0 |

LEGEND: R = Read only; -*n* = value after reset

**Table 18-18. POM Peripheral ID 5 Register (POMPERIPHERALID5) Field Descriptions**

| Bit | Field | Value | Description |
|---|---|---|---|
| 31-0 | Reserved | 0 | Reads return 0, writes have no effect. |

### 18.3.18 POM Peripheral ID 6 Register (POMPERIPHERALID6)

This is a CoreSight register and is for debug purpose only. This register reads 00000000.

**Figure 18-20. POM Peripheral ID 6 Register (POMPERIPHERALID6) [address = FFA0 4FD8h]**

| 31 … 0 |
|---|
| Reserved |
| R-0 |

LEGEND: R = Read only; -*n* = value after reset

**Table 18-19. POM Peripheral ID 6 Register (POMPERIPHERALID6) Field Descriptions**

| Bit | Field | Value | Description |
|---|---|---|---|
| 31-0 | Reserved | 0 | Reads return 0, writes have no effect. |

### 18.3.19 POM Peripheral ID 7 Register (POMPERIPHERALID7)

This is a CoreSight register and is for debug purpose only. This register reads 00000000.

**Figure 18-21. POM Peripheral ID 7 Register (POMPERIPHERALID7) [address = FFA0 4FDCh]**

| 31 … 0 |
|---|
| Reserved |
| R-0 |

LEGEND: R = Read only; -*n* = value after reset

**Table 18-20. POM Peripheral ID 7 Register (POMPERIPHERALID7) Field Descriptions**

| Bit | Field | Value | Description |
|---|---|---|---|
| 31-0 | Reserved | 0 | Reads return 0, writes have no effect. |

### 18.3.20 POM Peripheral ID 0 Register (POMPERIPHERALID0)

This is a CoreSight register and is for debug purpose only. This register shows the peripheral identification of the CoreSight component.

**Figure 18-22. POM Peripheral ID 0 Register (POMPERIPHERALID0) [address = FFA0 4FE0h]**

| 31–16 |
|---|
| Reserved |
| R-0 |

| 15–8 | 7–0 |
|---|---|
| Reserved | Part Number |
| R-0 | R-0 |

LEGEND: R/W = Read/Write; R = Read only; -*n* = value after reset

**Table 18-21. POM Peripheral ID 0 Register (POMPERIPHERALID0) Field Descriptions**

| Bit | Field | Value | Description |
|---|---|---|---|
| 31-8 | Reserved | 0 | Reads return 0, writes have no effect. |
| 7-0 | Part Number | 0 | Reads 0, since POMREV defines the module. |

### 18.3.21 POM Peripheral ID 1 Register (POMPERIPHERALID1)

This is a CoreSight register and is for debug purpose only. This register shows the peripheral identification of the CoreSight component.

**Figure 18-23. POM Peripheral ID 1 Register (POMPERIPHERALID1) [address = FFA0 4FE4]**

| 31–16 |
|---|
| Reserved |
| R-0 |

| 15–8 | 7–4 | 3–0 |
|---|---|---|
| Reserved | JEP106 Identity Code | Part Number |
| R-0 | R-7h | R-0 |

LEGEND: R/W = Read/Write; R = Read only; -*n* = value after reset

**Table 18-22. POM Peripheral ID 1 Register (POMPERIPHERALID1) Field Descriptions**

| Bit | Field | Value | Description |
|---|---|---|---|
| 31-8 | Reserved | 0 | Reads return 0, writes have no effect. |
| 7-4 | JEP106 Identity Code | 7h | Part of TI JEDEC number. |
| 3-0 | Part Number | 0 | Reads 0, since POMREV defines the module. |

### 18.3.22 POM Peripheral ID 2 Register (POMPERIPHERALID2)

This is a CoreSight register and is for debug purpose only. This register shows the peripheral identification of the CoreSight component.

**Figure 18-24. POM Peripheral ID 2 Register (POMPERIPHERALID2) [address = FFA0 4FE8h]**

| 31–16 |
|---|
| Reserved |
| R-0 |

| 15–8 | 7–4 | 3 | 2–0 |
|---|---|---|---|
| Reserved | Revision | JEDEC | JEP106 Identity Code |
| R-0 | R-0 | R-1h | R-1h |

LEGEND: R/W = Read/Write; R = Read only; -*n* = value after reset

**Table 18-23. POM Peripheral ID 2 Register (POMPERIPHERALID2) Field Descriptions**

| Bit | Field | Value | Description |
|---|---|---|---|
| 31-8 | Reserved | 0 | Reads return 0, writes have no effect. |
| 7-4 | Revision | 0 | Reads 0, since POMREV defines the module. |
| 3 | JEDEC | 1h | Indicates JEDEC assigned value. |
| 2-0 | JEP106 Identity Code | 1h | JEDEC+JEP106 Identity Code (POMPERIPHERALID2)+JEP106 Identity Code (POMPERIPHERALID1) form TI JEDEC ID of 97h. |

### 18.3.23 POM Peripheral ID 3 Register (POMPERIPHERALID3)

This is a CoreSight register and is for debug purpose only. This register reads 00000000.

**Figure 18-25. POM Peripheral ID 3 Register (POMPERIPHERALID3) [address = FFA0 4FECh]**

| 31–0 |
|---|
| Reserved |
| R-0 |

LEGEND: R = Read only; -*n* = value after reset

**Table 18-24. POM Peripheral ID 3 Register (POMPERIPHERALID3) Field Descriptions**

| Bit | Field | Value | Description |
|---|---|---|---|
| 31-0 | Reserved | 0 | Reads return 0, writes have no effect. |

### 18.3.24 POM Component ID 0 Register (POMCOMPONENTID0)

This is a CoreSight register and is for debug purpose only.

**Figure 18-26. POM Component ID 0 Register (POMCOMPONENTID0) [address = FFA0 4FF0h]**

| 31–16 |
|---|
| Reserved |
| R-0 |

| 15–8 | 7–0 |
|---|---|
| Reserved | Preamble |
| R-0 | R-Dh |

LEGEND: R/W = Read/Write; R = Read only; -*n* = value after reset

**Table 18-25. POM Component ID 0 Register (POMCOMPONENTID0) Field Descriptions**

| Bit | Field | Value | Description |
|---|---|---|---|
| 31-8 | Reserved | 0 | Reads return 0, writes have no effect. |
| 7-0 | Preamble | Dh | Preamble |

### 18.3.25 POM Component ID 1 Register (POMCOMPONENTID1)

This is a CoreSight register and is for debug purpose only.

**Figure 18-27. POM Component ID 1 Register (POMCPOMPONENTID1) [address = FFA0 4FF4h]**

| 31–16 |
|---|
| Reserved |
| R-0 |

| 15–8 | 7–4 | 3–0 |
|---|---|---|
| Reserved | Component Class | Preamble |
| R-0 | R-9h | R-0 |

LEGEND: R/W = Read/Write; R = Read only; -*n* = value after reset

**Table 18-26. POM Component ID 1 Register (POMCOMPONENTID1) Field Descriptions**

| Bit | Field | Value | Description |
|---|---|---|---|
| 31-8 | Reserved | 0 | Reads return 0, writes have no effect. |
| 7-4 | Component Class | 9h | CoreSight Component |
| 3-0 | Preamble | 0 | Preamble |

### 18.3.26 POM Component ID 2 Register (POMCOMPONENTID2)

This is a CoreSight register and is for debug purpose only.

**Figure 18-28. POM Component ID 2 Register (POMCPOMPONENTID2) [address = FFA0 4FF8h]**

| 31–16 |
|---|
| Reserved |
| R-0 |

| 15–8 | 7–0 |
|---|---|
| Reserved | Preamble |
| R-0 | R-5h |

LEGEND: R/W = Read/Write; R = Read only; -*n* = value after reset

**Table 18-27. POM Component ID 2 Register (POMCOMPONENTID2) Field Descriptions**

| Bit | Field | Value | Description |
|---|---|---|---|
| 31-8 | Reserved | 0 | Reads return 0, writes have no effect. |
| 7-0 | Preamble | 5h | Preamble |

### 18.3.27 POM Component ID 3 Register (POMCOMPONENTID3)

This is a CoreSight register and is for debug purpose only.

**Figure 18-29. POM Component ID 3 Register (POMCPOMPONENTID3) [address = FFA0 4FFCh]**

| 31–16 |
|---|
| Reserved |
| R-0 |

| 15–8 | 7–0 |
|---|---|
| Reserved | Preamble |
| R-0 | R-B1h |

LEGEND: R/W = Read/Write; R = Read only; -*n* = value after reset

**Table 18-28. POM Component ID 3 Register (POMCOMPONENTID3) Field Descriptions**

| Bit | Field | Value | Description |
|---|---|---|---|
| 31-8 | Reserved | 0 | Reads return 0, writes have no effect. |
| 7-0 | Preamble | B1h | Preamble |